

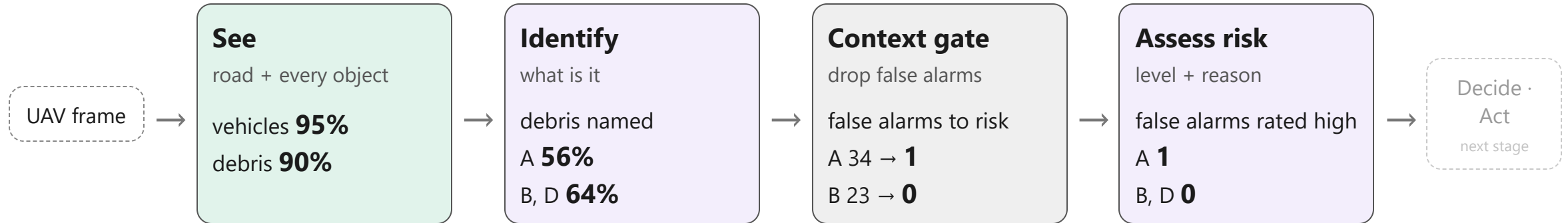
STAGE REPORT · 28 SEPTEMBER 2026

UAV-VisualSurv

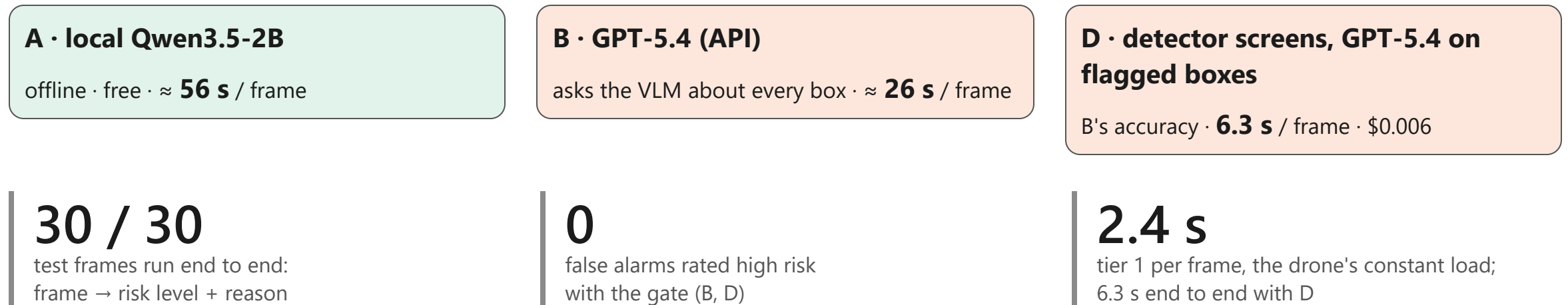
Highway hazard perception from a drone:
connecting the chain from seeing to risk, and comparing three architectures

This stage: connect the chain, compare the architectures

① Chain connected end to end · all 30 test frames



② Three architectures compared



Real UAV motorway frames + rendered 3D debris



Background

real UAV video, Pexels clip 8742752
Pexels License · open motorway only

Debris (red)

3D model "Low-Poly Treaded Tire"
by TepidGames · CC BY · Sketchfab
Blender 5.2.2 · random pose
real-world scale from calibrated
ground resolution (px per m)

Debris label

pixel-exact, from the render's alpha

Vehicle labels (blue)

Grounding DINO proposals
every box checked by hand

4 open-motorway locations

30 test frames · fixed

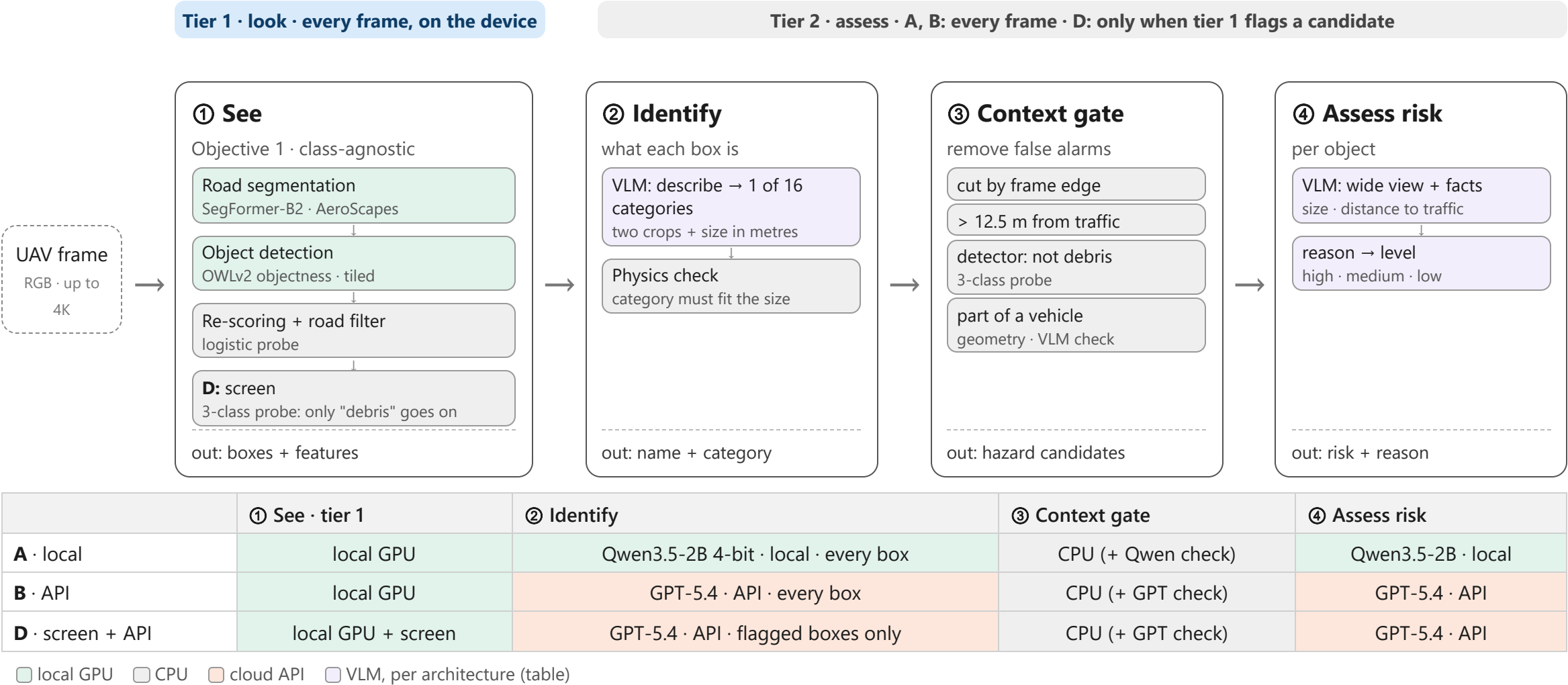
30 dev frames · all tuning

342 vehicles · hand-checked

1 debris object per test frame

leave-one-location-out scoring

One chain, three architectures

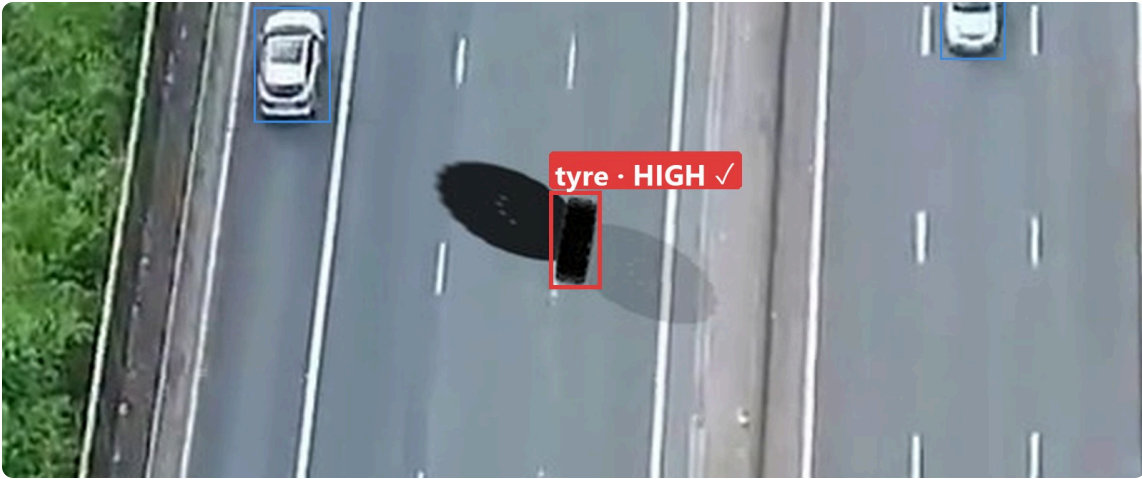


A · local Qwen3.5-2B

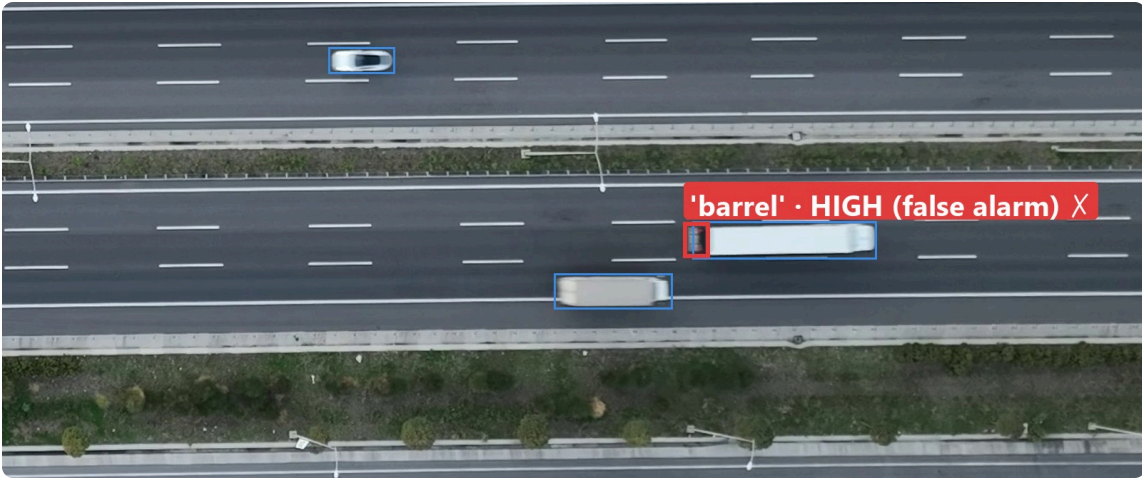
high medium removed by gate missed vehicle



✓ tyre found · HIGH · truck cut by frame edge removed



✓ small tyre found · HIGH



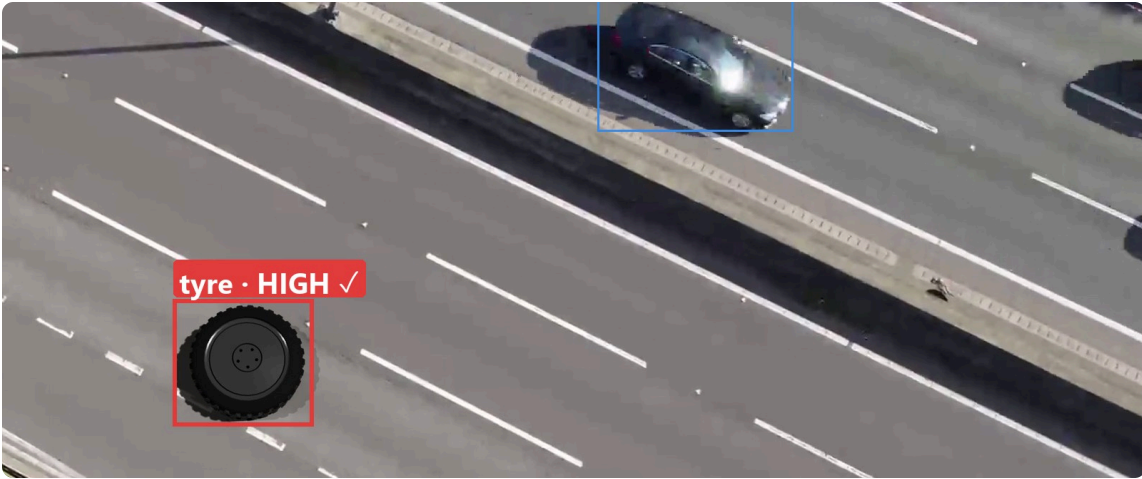
X lorry cab → 'barrel' · HIGH · false alarm reaches the alert



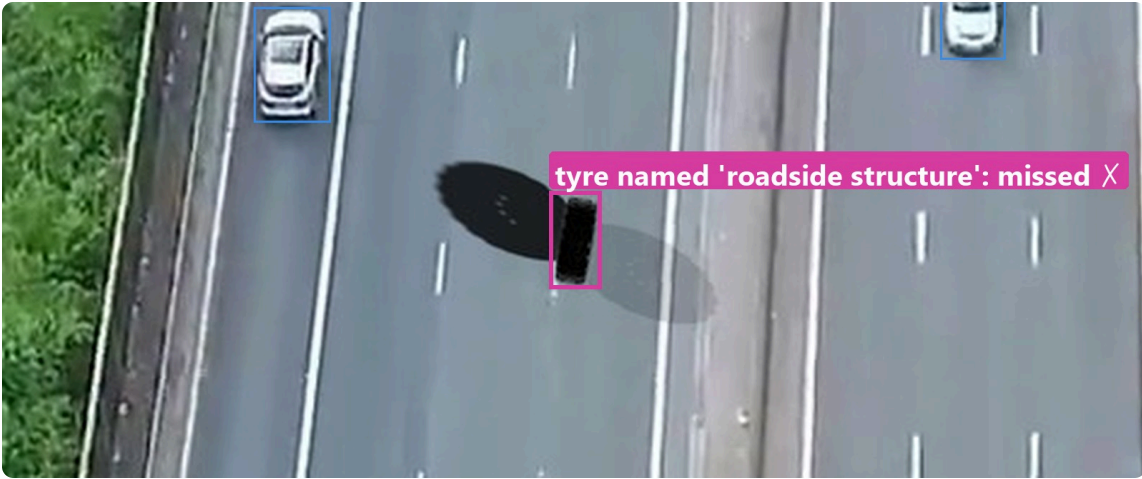
X fridge → 'roadside structure' · hazard missed

B · GPT-5.4

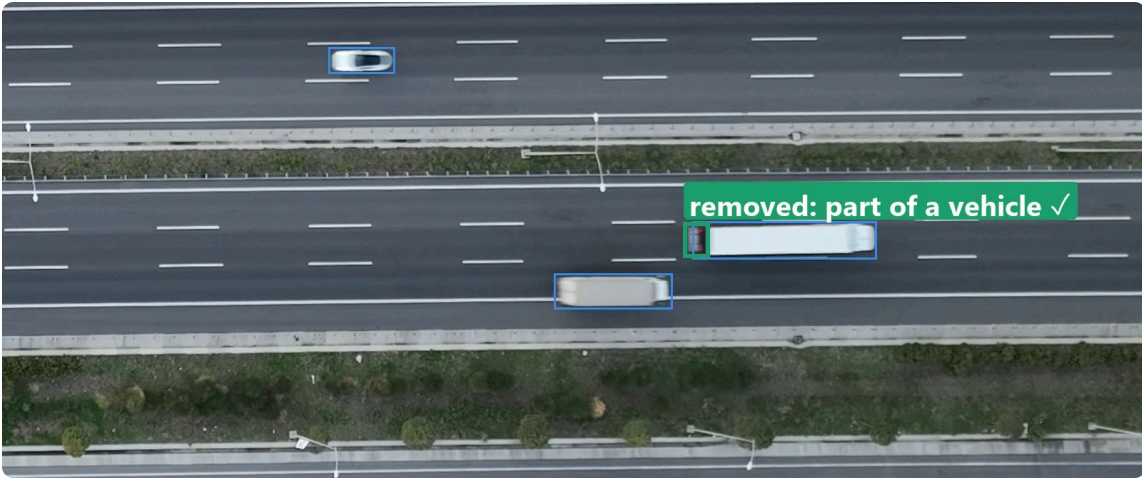
high medium removed by gate missed vehicle



✓ tyre found · HIGH



X small tyre → 'roadside structure' · missed



✓ lorry cab removed · GPT: "part of the vehicle"



✓ fridge found · HIGH · but it lies on the shoulder

D · detector screens first, GPT-5.4 on flagged boxes

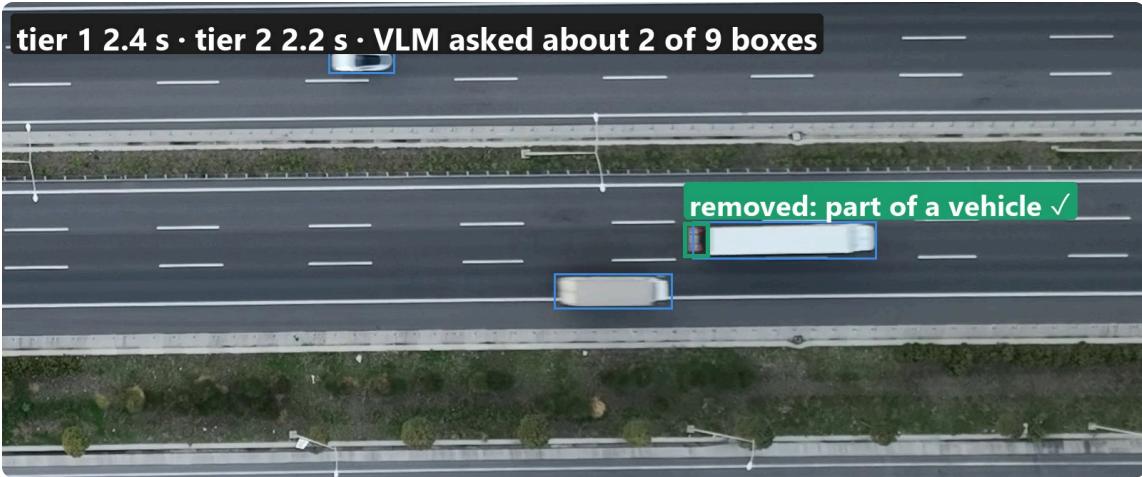
■ high
 ■ removed by gate
 ■ missed
 ■ vehicle (by the detector)
 · banner: this frame's tier times



✓ tyre found · HIGH · 1 of 5 boxes sent to the VLM



✗ small tyre → 'roadside structure' · as B · 1 of 30 boxes sent



✓ lorry cab removed · as B · 2 of 9 boxes sent



✓ fridge found · HIGH · 1 of 3 boxes sent

Time, accuracy and GPU in one table

	A · local Qwen	B · GPT-5.4	D · screen + GPT-5.4
Tier 1 · look · every frame, on the drone's GPU · s per frame, 30 test frames, RTX 3050 Ti (4 GB)			
road segmentation object detection (OWLv2), run side by side	2.36	2.36	2.36
re-scoring (+ screen in D)	< 0.01	< 0.01	< 0.01
TIER 1 TOTAL · can the drone keep up?	2.4	2.4	2.4
Tier 2 · assess · A, B: every frame · D: only frames tier 1 flags (27 of 30 here, all hold debris)			
identification (VLM) · bottleneck	49.2	22.0	2.1
context gate	0.09	0.13	0.05
risk assessment (VLM)	4.45	≈ 1.3	1.8
tier 2 total	53.7	≈ 23.4	4.0
VLM calls per frame	16.2	14.9	2.3
END TO END, s per frame	≈ 56	≈ 26	6.3
Accuracy · 30 test frames			
detection: vehicles · debris · precision	95.2% · 90.0% · 84.1% (shared)		
debris named correctly	56%	64%	64%
vehicles recognised	96.0% (VLM)	96.4% (VLM)	98.5% (detector)
debris boxes assessed for risk (of 27) · real debris lost by gate or screen	22 · 0	21 · 0	21 · 0
false alarms rated high	1	0	0
GPU memory · peak			
on the device	2.19 GB tier 1 · 1.97 GB VLM	2.19 GB	2.31 GB
all models loaded at once	4.05 GB > 4 GB card	2.19 GB	2.31 GB
Deployment			
API cost per frame · runs offline	\$0 · yes	\$0.036 · no	\$0.0056 · no

Where we stand, what comes next

This stage

- ✓ chain connected: frame → objects → identity → gate → risk + reason
- ✓ "no false alarm rated high": B and D on the test set
- ✓ D: B's accuracy at 6.3 s per frame · tier 1 2.4 s
- A: offline and free, but ≈ 56 s per frame

Next stage

- decide + act: lane signals, message signs
- tier 1 below 1 s, without losing detections
- small-object detection (pallets ≈ 22 px)
- object size from altitude + focal length